

**TECHNICAL SPECIFICATION FOR TENSION HARDWARE FITTINGS COMPRISING OF CROSS-ARM STRAP, BALL CLEVIS & SOCKET TONGUE AND BOLTED TYPE TENSION CLAMP (COMPOSITE HARDWARE FITTINGS) TO BE USED FOR ACSR WEASEL/RABBIT CONDUCTORS (30/50 SQ MM.) & ACSR DOG CONDUCTOR (100 SQ MM)**

**1.0 SCOPE :**

- The Specification covers the design, manufacture, testing (preferably at manufacturer's works before supply) and delivery of the **combined unit of Cross Arm Strap, Ball Clevis & Socket Tongue and bolted type Tension Clamp (Composite Hardware Fittings)** including loading, unloading anywhere in West Bengal.
- The combined units offered shall be complete with all components which are necessary (excepting disc insulator) or usual for their effective performance and easy maintenance and inter changeability at site. Such parts shall be deemed to be within the scope of contract.

**2.0 GENERAL REQUIREMENTS:**

- Ball designation should be **16 mm**.
- Hardware Fittings with **ultimate tensile strength on aluminium tension clamp should be of 30 KN (minimum)** and **ultimate tensile strength on steel parts assembly should be of 45 KN (minimum)**, should be used for **ACSR Weasel/Rabbit** conductors.  
Hardware Fittings with **ultimate tensile strength of 45 KN** should be used for **ACSR Dog** conductors.
- The Fittings shall be free from defects, corrosion protected and shall meet the requirements of Galvanizing Test etc. as per relevant IS.
- All forging and casting shall be of good finish and free from flaws and other defects. The edges of the fittings such as the tongue, clevis and holes shall be rounded.
- All parts of different fittings which provide the interconnection shall be made such that sufficient clearance is provided at the connection point to ensure free movement. All tongue and clevis connection shall be free in this manner but care shall be taken that too much clearance for the tongue & clevis is avoided.
- Spring washer should be electro galvanized-Coating thickness as per **IS : 1573-1986**.

**U bolt, Hexagonal Bolt, Nut, Plain washer and all other ferrous parts shall be Hot dip Galvanized.**  
**In case of Hot Dip Galvanization, minimum value or average mass of Zinc coating (g/m<sup>2</sup>) should be as per Table 1 of IS 4759 for different classes of materials.**

**The Split Pin/Cotter Pin to be used shall be Humpback type & shall be made of Stainless Steel conforming to IS: 5522-1992 with a minimum hardness of 160HV.**

**Locking device (R type/ W Clip) for Ball and Socket Lockers shall be of Stainless Steel conforming to IS 6603-1972 with minimum hardness of 160HV. The dimension shall conform to IS:2486(Part 3)-1974.**

**3.0 STANDARD**

The Combined units shall conform to the provisions of Indian Standards relevant to each individual component except where specified otherwise.

The following I.S. amended upto date shall be the guideline for manufacture & testing.

|      |                                                                         |   |                                                                                                                   |
|------|-------------------------------------------------------------------------|---|-------------------------------------------------------------------------------------------------------------------|
| i)   | <b>IS 2486 ( Part 1 ) : 1993</b>                                        | : | Specification for metal fittings of insulators for overhead power lines with nominal voltage greater than 1000 V. |
| ii)  | <b>IS 2486 ( Part 2 ) : 1989</b><br><b>IS : 2486( Part III ) - 1974</b> | : | Specification for Insulator fittings for overhead power lines with nominal voltage greater than 1000 V.           |
| iii) | <b>IS 4759 : 1996</b>                                                   | : | <b>Specification for hot-dip zinc coatings on structural steel and other allied products.</b>                     |
| iv)  | <b>IS : 6745 - 1872</b>                                                 | : | Determination of mass of zinc coating on zinc coated iron and steel articles.                                     |
| v)   | <b>IS : 2633 - 1986</b>                                                 | : | <b>Method for testing uniformity of coating on zinc coated.</b>                                                   |
| vi)  | <b>IS:1573 - 1986</b>                                                   | : | Specification for electroplated coatings of zinc on iron and steel.                                               |

#### 4.0 MATERIALS AND OTHER REQUIREMENTS :

##### 4.1: CROSS-ARM STRAPS (fig-1):

Cross-arm straps shall be manufactured from **MS Flat hot dip galvanized** and to connect the cross-arm/bracket of the structure at one end and the Ball Clevis at the other end.

It should be complete with **hexagonal bolts, nuts, spring washers and Cotter pin at the threaded end to lock the unit.**

Minimum Threaded portion of the bolt shall be **30mm.**

| CROSS-ARM STRAPS to be used for | Ultimate tensile strength(minimum) |
|---------------------------------|------------------------------------|
| ACSR Weasel/Rabbit Conductor    | 45 KN                              |

Dimensions shall be in accordance with **IS: 2486 (Part-2)** unless otherwise specified.

##### 4.2: BALL CLEVIS (fig-2) & SOCKET TONGUE(fig-3):

All **forgings** shall be of good finish and free from any flaws and any other defects which may cause decrement of efficiency while in operation. Connection/attachment with other component of the unit shall provide reasonable clearance/ensure free movement at the connecting/attaching point. Care should be taken to avoid too much clearance while used with insulators.

Materials for Ball Clevis & Socket Tongue should be as stated below:-

**Ball clevis - Forged Steel and hot dipped galvanized.**  
**Socket Tongue - Forged Steel and hot dipped galvanized.**

| BALL CLEVIS & SOCKET TONGUE to be used for | Ultimate tensile strength(minimum) |
|--------------------------------------------|------------------------------------|
| ACSR Weasel/Rabbit Conductor               | 45 KN                              |

All dimensions of Ball Clevis & Socket Tongue shall be in accordance with **IS: 2486 (Part-2)** unless otherwise specified.

##### 4.3 Bolted type Tension Clamp :

**Fig 4 is for Tension Hardware Fittings suitable for ACSR Weasel/Rabbit Conductor and Fig 5 is for Tension Hardware Fittings suitable for ACSR Dog Conductor.**

All **forgings & castings** shall be of good finish and free from flaws or any other defects which may cause decrement of efficiency while in operation. The edges on the outside of the fittings and the grooves shall be smooth & rounded. Sharp radius of curvature, ridges etc. which may lead to localized pressure or cause damage to the conductors in service shall be avoided. The clamp shall permit the conductor to slip before the failure of conductor occurs.

**Tension clamps shall be bolted type.**

Tension clamps for ACSR Weasel/ Rabbit conductor shall be single bolted type with minimum one no. keeper. Ultimate tensile strength of Tension clamps should not be less than 30 KN for ACSR Weasel/Rabbit Conductor.

Tension clamps for ACSR Dog conductor shall be three bolted type with minimum one no. keeper. Ultimate tensile strength of Tension clamps should not be less than 45 KN for ACSR Dog conductor.

The material for Tension clamps & keeper should be high strength Aluminum Alloy.

The slip strength should not be less than **95% UTS** of the respective conductor.

Tension clamps shall be fitted with **rivet, flat washer, cotter pin, 'U' bolts & nuts, flat washers & spring washers.**



4.4 Fig 6 is for Composite Hardware Fittings of Insulators comprising of Cross arm straps, ball clevis and socket tongue ,bolted type tension clamp along with all nuts, bolts, washers ,security clip, split/cotter pins , suitable for ACSR Weasel/Rabbit Conductor.

Minimum gross weight of complete fittings (except Insulator) of Composite Hardware Fittings to be procured for ACSR Weasel/Rabbit Conductor should be 2.00kg.

Fig 7 is for Composite Hardware Fittings of insulators comprising of Cross arm straps, ball clevis and socket tongue ,bolted(three bolted) type tension clamp along with all nuts, bolts, washers ,security clip, split/cotter pins , suitable for ACSR Dog Conductor.

Minimum gross weight of complete fittings (except Insulator) of Composite Hardware Fittings to be procured for ACSR Dog Conductor should be 2.3 kg.

## 5.0 MARKING:

Tension Hardware Fittings suitable for ACSR Weasel/Rabbit Conductor should be supplied in a string.

Tension Clamp shall be legibly & indelibly marked with Name or Trade mark of manufacturer, year of manufacture, **30 KN** and suitable identification marking of the property 'WBSEDCL'.

Tension Hardware Fittings suitable for ACSR Dog Conductor should be supplied in a string.

Tension Clamp shall be legibly & indelibly marked with Name or Trade mark of manufacturer, year of manufacture, **45 KN** and suitable identification marking of the property 'WBSEDCL'.

The marking should be engraved /embossed.

## 6.0 TESTS AND TEST CERTIFICATES :

### Type Test :

Along with the bid, the bidder has to submit Type Test Reports on same fittings carried out within last five years from the date of opening of techno-commercial bid of the tender from any NABL accredited Laboratory having NABL logo. Otherwise the tender may be rejected.

The following shall constitute the Type Test for Tension Clamps

1. Visual Examination
2. Verification of dimensions
3. Mechanical Test (Slip Strength Test)
4. Ultimate Tensile Strength Test
5. Galvanizing / Electroplating Test (for ferrous parts)
6. Chemical Composition Test

The following shall constitute Type Tests on tension hardware fittings except Clamp:

1. Visual Examination
2. Verification of dimensions
3. Mechanical Test
4. Galvanizing Test
5. Chemical Composition Test

### ACCEPTANCE TEST

The following shall constitute acceptance test for Clamps:

1. Visual Examination
2. Verification of dimensions
3. Mechanical Test (Slip Strength Test)
4. Ultimate Tensile Strength Test
5. Galvanizing Test (for ferrous parts)
6. Chemical Composition Test

Acceptance Test for tension hardware fittings except Clamp

1. Visual Examination
2. Verification of dimensions
3. Ultimate Tensile Strength Test
4. Galvanizing Test
5. Chemical Composition Test

## 7.0 TOLERANCE :

Tolerance shall be as per IS: 2486 (Part-I).

## 8.0 GUARANTEE:

In the event of any defect in the equipment/materials arising out of faulty design, materials, workmanship within a period of 12 (twelve) months of commissioning or 18 (eighteen) months from the date of last despatch of any integral part of the equipment/materials whichever is earlier, the supplier shall guarantee to replace or repair the same to the satisfaction of the purchaser.

If the supplier fail to do so within a reasonable time, WBSEDCL reserves the right to effect repair or replacement by any other agency and recover charges for repair or replacement from the supplier.

## GUARANTEED TECHNICAL PARTICULARS

TENSION HARDWARE FITTINGS COMPRISING OF CROSS-ARM STRAP, BALL CLEVIS & SOCKET TONGUE AND BOLTED TYPE TENSION CLAMP TO BE USED FOR ACSR WEASEL/RABBIT CONDUCTORS

|     |                                                                                     |                                                                              |
|-----|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| 1.  | Name & Address of the Firm                                                          |                                                                              |
| 2.  | Type of Insulator Fittings                                                          | B&S type                                                                     |
| 3.  | Ball Designation                                                                    | 16 mm                                                                        |
| 4.  | Ultimate Tensile Strength(Minimum)                                                  |                                                                              |
| a)  | Bolted Type (Single Bolt) Tension Clamp                                             | 30 KN                                                                        |
| b)  | Complete Fittings                                                                   | 30 KN                                                                        |
| 5.  | Whether dimension other than specified are in accordance to IS: 2486 (Part-II)-1974 |                                                                              |
| 6.  | Material used                                                                       |                                                                              |
| a)  | Cross arm Strap                                                                     | Mild Steel, hot dip galvanized                                               |
| b)  | Ball Clevis                                                                         | Forged Steel and hot dipped galvanized                                       |
| c)  | Socket Tongue                                                                       | Forged Steel and hot dipped galvanized                                       |
| d)  | Bolted type Tension Clamp & keeper                                                  | High Strength Aluminium Alloy                                                |
| e)  | Locking devices for ball and socket lockers                                         | Stainless Steel conforming to IS: 6603-1972 with minimum hardness of 160 HV. |
| f)  | Split Pins/ Cotter pins                                                             | Stainless Steel conforming to IS: 5522-1992 with minimum hardness of 160 HV. |
| g)  | U-Bolts & Nut                                                                       | MS, Hot Dip Galvanized                                                       |
| h)  | Hexagonal Bolt & Nut                                                                | MS, Hot Dip Galvanized                                                       |
| i)  | Spring Washer                                                                       | Spring Steel, Electro Galvanized                                             |
| j)  | Plain Washer                                                                        | MS Hot Dip Galvanized                                                        |
| 7.  | Zinc conforming to grade                                                            | As per IS 209: 1992                                                          |
| 8.  | Weight per set                                                                      |                                                                              |
| a)  | Gross weight of complete fittings                                                   |                                                                              |
| b)  | Weight of Clamp & keeper                                                            |                                                                              |
| c)  | Weight of Cross Arm Strap                                                           |                                                                              |
| d)  | Weight of Ball Clevis                                                               |                                                                              |
| e)  | Weight of Socket Tongue                                                             |                                                                              |
| f)  | Weight of Nut, Bolts, Rivet, U-Bolt & accessories                                   |                                                                              |
| g)  | Minimum Guaranteed weight of Fittings                                               |                                                                              |
| h)  | Whether galvanizing confirm to IS-2633/1964                                         |                                                                              |
| 9.  | Packing details                                                                     |                                                                              |
| a)  | Net Weight                                                                          |                                                                              |
| b)  | Gross Weight                                                                        |                                                                              |
| c)  | Contents of each Pack                                                               |                                                                              |
| d)  | Type of Packing                                                                     |                                                                              |
| 10. | Tolerance in dimension, if any                                                      | IS:2486 Part I                                                               |
| 11. | Marking                                                                             |                                                                              |
| 12. | Guarantee                                                                           |                                                                              |



# GUARANTEED TECHNICAL PARTICULARS

**TENSION HARDWARE FITTINGS COMPRISING OF CROSS-ARM STRAP, BALL CLEVIS & SOCKET TONGUE AND BOLTED TYPE TENSION CLAMP TO BE USED FOR ACSR DOG CONDUCTOR**

|     |                                                                                     |                                                                              |
|-----|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| 1.  | Name & Address of the Firm                                                          |                                                                              |
| 2.  | Type of Insulator Fittings                                                          | B&S type                                                                     |
| 3.  | Ball Designation                                                                    | 16 mm                                                                        |
| 4.  | Ultimate Tensile Strength(Minimum)                                                  |                                                                              |
| a)  | Bolted Type (three bolted) Tension Clamp                                            | 45 KN                                                                        |
| b)  | Complete Fittings                                                                   | 45 KN                                                                        |
| 5.  | Whether dimension other than specified are in accordance to IS: 2486 (Part-II)-1974 |                                                                              |
| 6.  | Material used                                                                       |                                                                              |
| a)  | Cross arm Strap                                                                     | Mild Steel, hot dip galvanized                                               |
| b)  | Ball Clevis                                                                         | Forged Steel and hot dipped galvanized                                       |
| c)  | Socket Tongue                                                                       | Forged Steel and hot dipped galvanized                                       |
| d)  | Bolted type Tension Clamp & keeper                                                  | High Strength Aluminium Alloy                                                |
| e)  | Locking devices for ball and socket lockers                                         | Stainless Steel conforming to IS: 6603-1972 with minimum hardness of 160 HV. |
| f)  | Split Pins /Cotter pins                                                             | Stainless Steel conforming to IS: 5522-1992 with minimum hardness of 160 HV. |
| g)  | U-Bolts & Nut                                                                       | MS, Hot Dip Galvanized                                                       |
| h)  | Hexagonal Bolt & Nut                                                                | MS, Hot Dip Galvanized                                                       |
| i)  | Spring Washer                                                                       | Spring Steel, Electro Galvanized                                             |
| j)  | Plain Washer                                                                        | MS Hot Dip Galvanized                                                        |
| 7.  | Zinc conforming to grade                                                            | As per IS 209: 1992                                                          |
| 8.  | Weight per set                                                                      |                                                                              |
| a)  | Gross weight of complete fittings                                                   |                                                                              |
| b)  | Weight of Clamp & keeper                                                            |                                                                              |
| c)  | Weight of Cross Arm Strap                                                           |                                                                              |
| d)  | Weight of Ball Clevis                                                               |                                                                              |
| e)  | Weight of Socket Tongue                                                             |                                                                              |
| f)  | Weight of Nut, Bolts, Rivet, U-Bolt & accessories                                   |                                                                              |
| g)  | Minimum Guaranteed weight of Fittings                                               |                                                                              |
| h)  | Whether galvanizing confirm to IS-2633/1964                                         |                                                                              |
| 9.  | Packing details                                                                     |                                                                              |
| a)  | Net Weight                                                                          |                                                                              |
| b)  | Gross Weight                                                                        |                                                                              |
| c)  | Contents of each Pack                                                               |                                                                              |
| d)  | Type of Packing                                                                     |                                                                              |
| 10. | Tolerance in dimension, if any                                                      | IS:2486 Part I                                                               |
| 11. | Marking                                                                             |                                                                              |
| 12. | Guarantee                                                                           |                                                                              |

# DRAWING

## CROSS ARM STRAP:

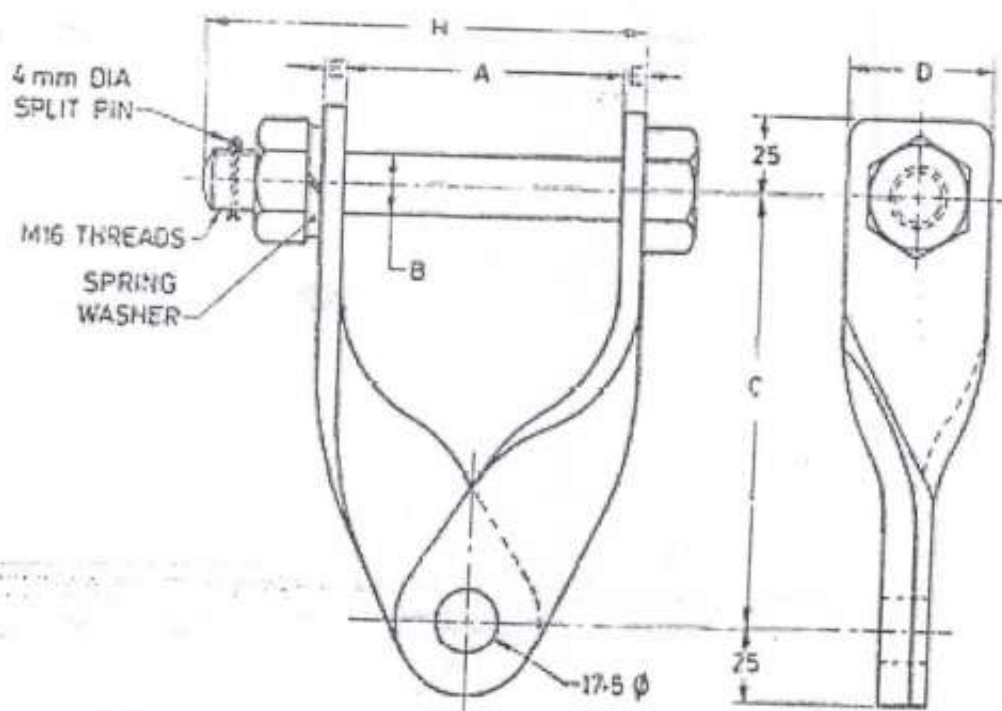


Fig-1: Cross Arm Strap

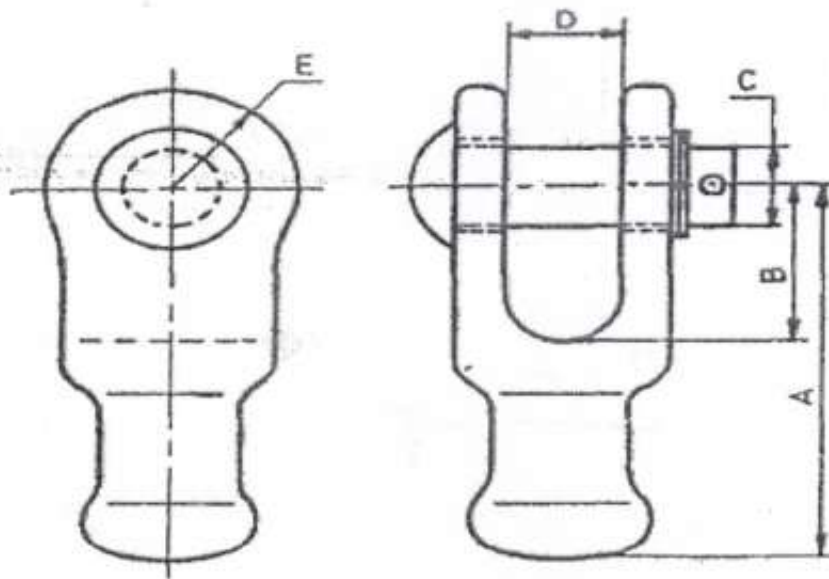
Dimensions as per IS 2486(Part II):

| A   | B (Dia) | C   | D  | E | H   |
|-----|---------|-----|----|---|-----|
| 100 | 16      | 140 | 35 | 6 | 145 |

(All dimensions are in mm)

| PARTICULARS         | MATERIAL                        | QTY.  |
|---------------------|---------------------------------|-------|
| TWISTED STRAP       | MILD STEEL- HOT DIP GALVANIZED  | 2 No. |
| M-16 X 145 MM BOLT  | MILD STEEL- HOT DIP GALVANIZED  | 1 No. |
| 16 MM SPRING WASHER | SPRING STEEL-ELECTRO GALVANIZED | 1 No. |
| 16MM NUT            | MILD STEEL- HOT DIP GALVANIZED  | 1 No. |
| SPLIT/COTTER PIN    | STAINLESS STEEL                 | 1 No. |

## BALL CLEVIS :



**Fig 2: Ball Clevis**

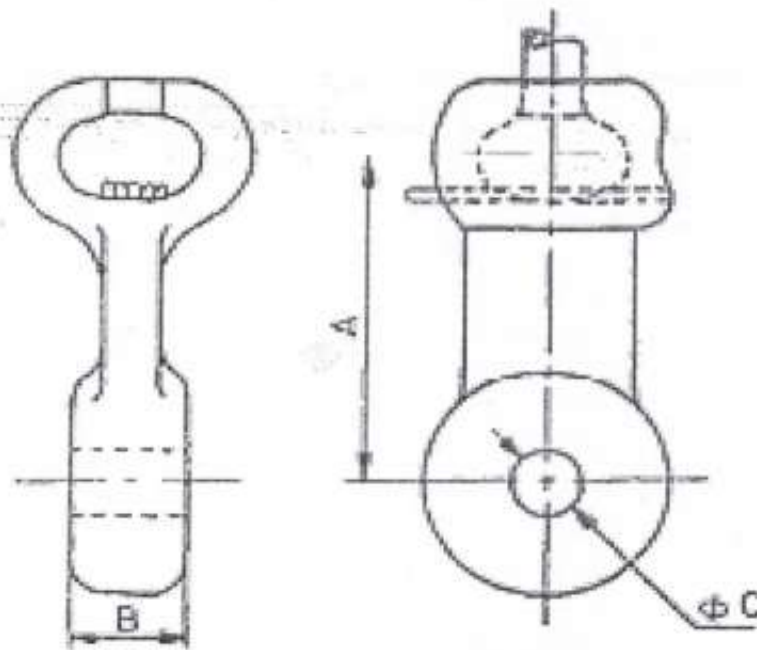
Dimensions as per IS 2486(Part II):

| A  | B  | C  | D  | E    | Pin Ball Designation |
|----|----|----|----|------|----------------------|
| 76 | 32 | 16 | 19 | 20.5 | 16                   |

(All dimensions are in mm)

| PARTICULARS               | MATERIAL                         | QTY.         |
|---------------------------|----------------------------------|--------------|
| <b>BALL CLEVIS</b>        | FORGED STEEL- HOT DIP GALVANIZED | <b>1 No.</b> |
| <b>16 MM RIVET</b>        | MILD STEEL- HOT DIP GALVANIZED   | <b>1 No.</b> |
| <b>16 MM PLAIN WASHER</b> | MILD STEEL- HOT DIP GALVANIZED   | <b>1 No.</b> |
| <b>SPLIT/COTTER PIN</b>   | STAINLESS STEEL                  | <b>1 No.</b> |

# SOCKET TONGUE:



**Fig-3: Socket Tongue**

**Dimensions as per IS 2486(Part II):**

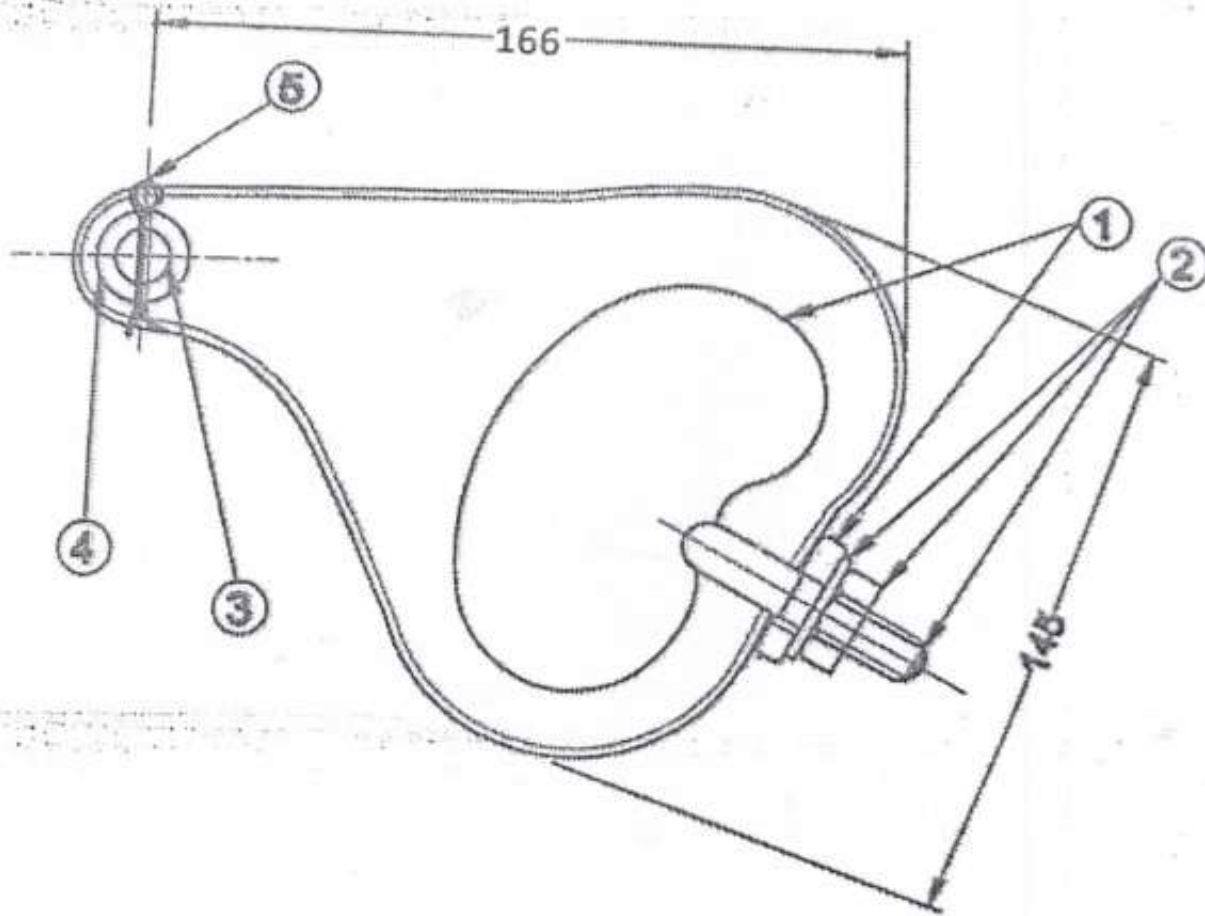
| A  | B  | C    | Pin Ball Designation |
|----|----|------|----------------------|
| 51 | 16 | 17.5 | 16                   |

(All dimensions are in mm)

| PARTICULARS          | MATERIAL                         | QTY.         |
|----------------------|----------------------------------|--------------|
| <b>SOCKET TONGUE</b> | FORGED STEEL- HOT DIP GALVANIZED | <b>1 No.</b> |
| <b>SECURITY CLIP</b> | STAINLESS STEEL                  | <b>1 No.</b> |



# SINGLE BOLTED TYPE TENSION CLAMP

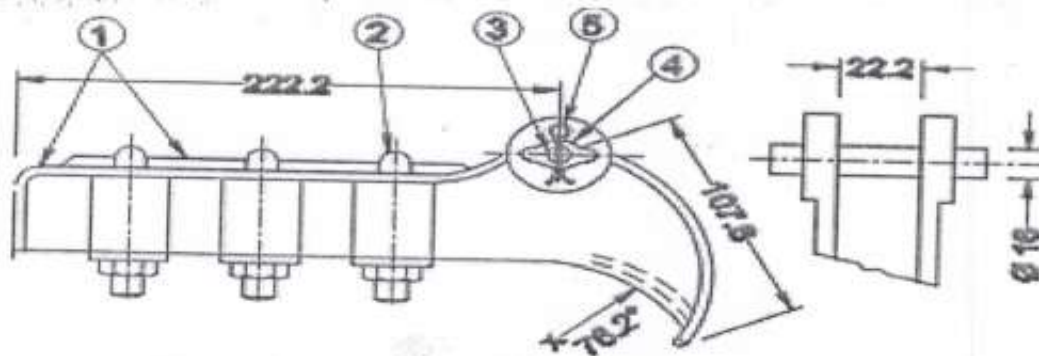


**Fig 4: BOLTED TYPE TENSION CLAMP for CHF of ACSR WEASEL/RABBIT CONDUCTOR**

Dimensions as per IS 2486 (Part II): All dimensions are in mm.

| SL No. | PARTICULARS              | MATERIAL                        | QTY.  |
|--------|--------------------------|---------------------------------|-------|
| 1.     | ALUMINIUM CLAMP & KEEPER | ALUMINIUM ALLOY                 | 1 SET |
| 2.     | U-BOLT                   | MILD STEEL- HOT DIP GALVANIZED  | 1 No. |
|        | NUT                      | MILD STEEL- HOT DIP GALVANIZED  | 2 No. |
|        | PLAIN WASHER             | MILD STEEL- HOT DIP GALVANIZED  | 2 No. |
|        | SPRING WASHER            | SPRING STEEL-ELECTRO GALVANIZED | 2 No. |
| 3.     | 16 MM RIVET              | MILD STEEL- HOT DIP GALVANIZED  | 1 No. |
| 4.     | 16 MM PLAIN WASHER       | MILD STEEL- HOT DIP GALVANIZED  | 1 No. |
| 5.     | SPLIT/COTTER PIN         | STAINLESS STEEL                 | 1 No. |

## Strain Clamp:



## ALUMINIUM TENSION CLAMP

| Material        | Type of Clamp | Conductor Dia |         | No. of U Bolts | No. of Keeper |
|-----------------|---------------|---------------|---------|----------------|---------------|
|                 |               | Minimum       | Maximum |                |               |
| Aluminium Alloy | A             | 7.6           | 17.8    | 3              | 1             |

(All dimensions are in mm)

Fig 5: Bolted Type Aluminium Tension Clamp suitable for ACSR Dog Conductor

| Sl. No. | Particulars                     | Material                        | Quantity      |
|---------|---------------------------------|---------------------------------|---------------|
| 1.      | <b>Tension Clamp and Keeper</b> | Aluminium Alloy                 | <b>1 set</b>  |
| 2.      | <b>U-Bolt</b>                   | Mild Steel-Hot Dip Galvanized   | <b>3 nos.</b> |
|         | <b>Nut</b>                      | Mild Steel-Hot Dip Galvanized   | <b>6 nos.</b> |
|         | <b>Plain Washer</b>             | Mild Steel-Hot Dip Galvanized   | <b>6 nos.</b> |
|         | <b>Spring Washer</b>            | Spring Steel-Electro Galvanized | <b>6 nos.</b> |
| 3.      | <b>16 mm Rivet</b>              | Mild Steel-Hot Dip Galvanized   | <b>1 no.</b>  |
| 4.      | <b>16 mm Plain Washer</b>       | Mild Steel-Hot Dip Galvanized   | <b>1 no.</b>  |
| 5.      | <b>Split/Cotter Pin</b>         | Stainless Steel                 | <b>1 no.</b>  |

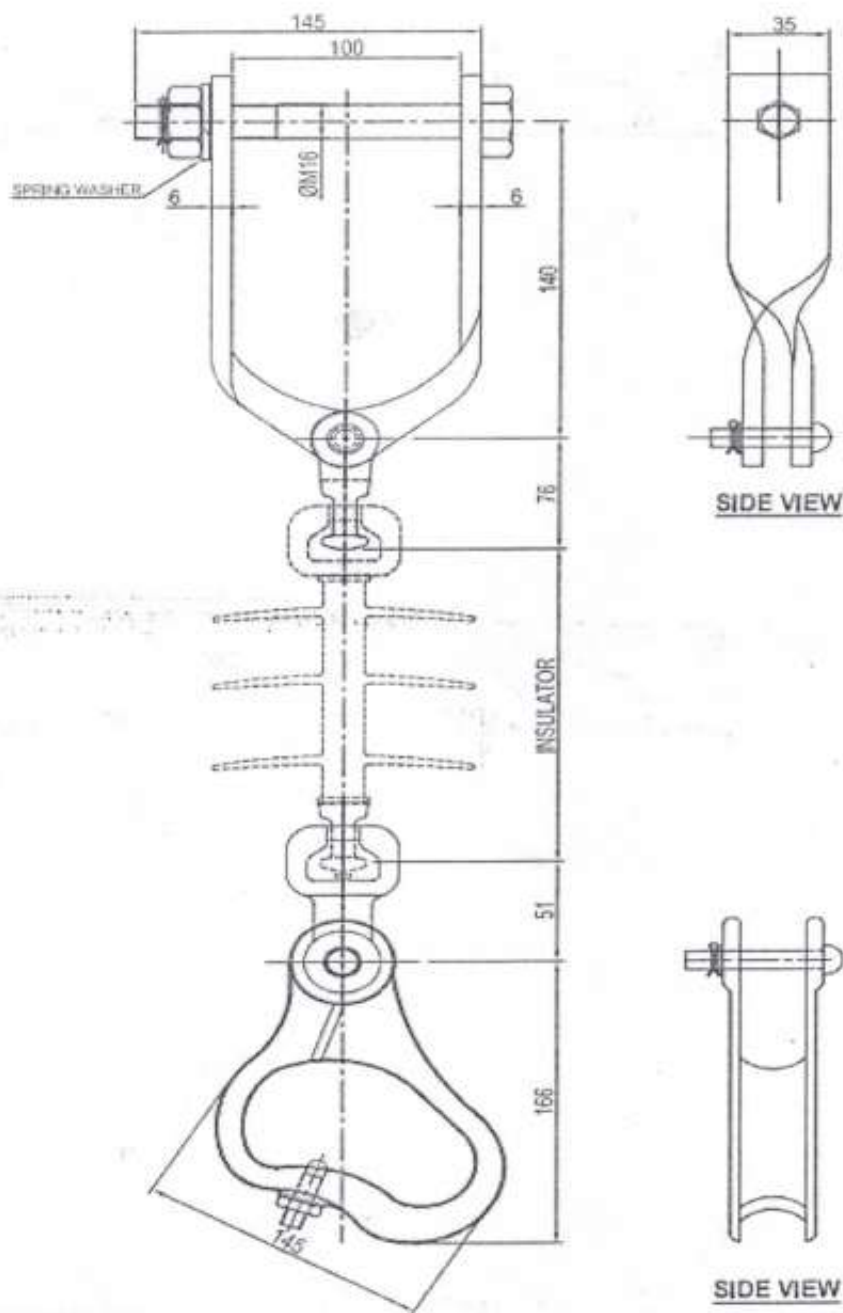


Fig. 6: Composite Hardware Fittings of insulators comprising of all Accessories to be used for ACSR Weasel/Rabbit Conductors.

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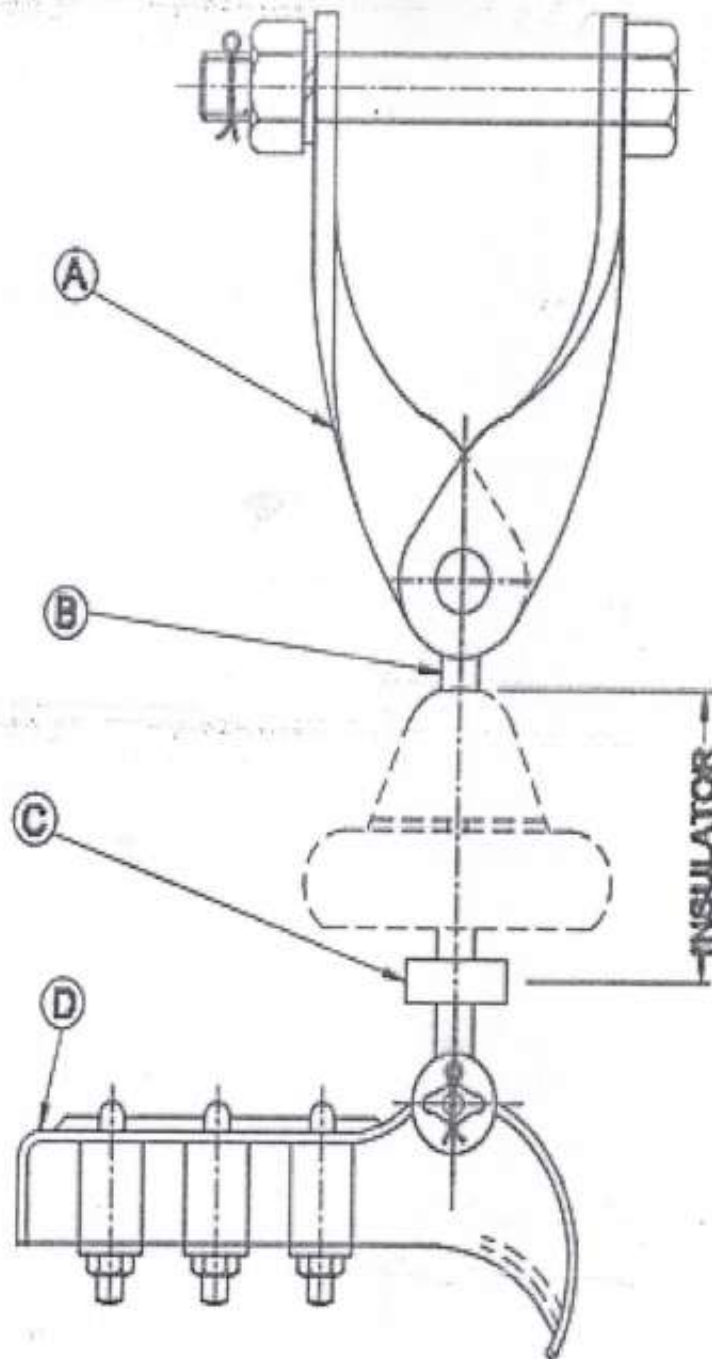
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| Mark | Particulars             |
|------|-------------------------|
| A    | CROSS ARM STRAP         |
| B    | BALL CLEVIS             |
| C    | SOCKET TONGUE           |
| D    | ALUMINIUM TENSION CLAMP |

Fig 7: Composite Hardware Fittings of Insulators comprising of all accessories to be used for ACSR Dog Conductors.